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Cooperative Search and Rescue with Drone Swarm



Luiz Giacomossi, Marcos R. O. A. Maximo, Nils Sundelius, Peter Funk, José F. B. Brancalion, and Rickard Sohlberg

Abstract Unmanned Aerial Vehicle (UAV) swarms, also known as drone swarms, have been a subject of extensive research due to their potential to enhance monitoring, surveillance, and search missions. Coordinating several drones flying simultaneously presents a challenge in increasing their level of automation and intelligence to improve strategic organization. To address this challenge, we propose a solution that uses hill climbing, potential fields, and search strategies in conjunction with a probability map to coordinate a UAV swarm. The UAVs are autonomous and equipped with distributed intelligence to facilitate a cooperative search application. Our results show the effectiveness of the swarm, indicating that this approach is a promising approach to addressing this problem.

Keywords Drones · UAV · Search and rescue · Swarm · Cooperative

L. Giacomossi (✉) · M. R. O. A. Maximo
Autonomous Computational Systems Lab (LAB-SCA), Aeronautics Institute of Technology (ITA), São José Dos Campos, Brazil
e-mail: luiz.giacomossi@ga.ita.br

M. R. O. A. Maximo
e-mail: mmaximo@ita.br

N. Sundelius · P. Funk · R. Sohlberg
Mälardalen University (MDU), Universitetsplan 1, 721 23 Västerås, Sweden
e-mail: nils.sundelius@mdu.se

P. Funk
e-mail: peter.funk@mdu.se

R. Sohlberg
e-mail: rickard.sohlberg@mdu.se

J. F. B. Brancalion
Technological Development Department, EMBRAER S.A, São José Dos Campos, Brazil
e-mail: jose.brancalion@embraer.com.br

1 Introduction

Swarms are a natural and widespread phenomenon among various groups of animals. They are able to move in space with coordination and cohesion through simple rules. In the context of unmanned aerial vehicles (UAVs), swarms refer to a group of homogeneous entities working collaboratively towards a common goal [1]. This concept has drawn the interest of researchers due to its potential for efficient and cost-effective completion of complex tasks in harsh environments compared to traditional single drone applications. This approach aims to achieve the highest level of autonomy, which poses several challenges [2]. One of these challenges is the ability to perform a task coordinated among multiple UAVs without human operator intervention [3].

Using UAVs in search and rescue operations (SAR) can significantly improve the chances of locating and rescuing individuals in distress. UAVs equipped with advanced sensors and cameras can cover large areas quickly, allowing search teams to identify potential targets with greater speed and efficiency. They can also access hard-to-reach or dangerous areas, providing a high level view of the situation and allowing rescuers to assess the situation more accurately. In addition, UAVs can reduce the risk to human search teams, allowing them to focus on the most critical areas of a search and minimizing their exposure to danger. Recent research has shown the potential of UAVs in SAR. Dantas et al. [5] explore low-processing image recognition alternatives, such as Inception and Yolo, to address challenges such as locating small targets in large areas. Sampedro et al. [14] propose a real-time mission planning architecture for a swarm of UAVs with dynamic agent-to-task assignment, while Volchenkov et al. [15] propose path planning methods using fuzzy logic and PSO algorithms.

Mission planning involves developing a strategy to accomplish an objective in a specific mission. It includes identifying required resources, risks, and challenges to develop a detailed plan with tasks and responsibilities. Mission planning at an online level requires constant updates as environmental conditions change. Planning is one of the biggest challenges in artificial intelligence (AI), and building intelligent systems that can plan effectively often requires a hybrid approach. There is no single algorithm that can perform perfect planning, and the challenge is to develop planning approaches that are as efficient as possible for a specific task. As humans, we often begin by making a coarse-grained plan, even if we do not have a complete understanding of the task and its challenges. At the top level, the task for the coarse-grained plan may have conflicting goals, such as minimizing the risk of problems, finding the target as quickly as possible, and using as few resources as possible. Humans perform online planning by continuously reevaluating and adjusting our plans as the situation changes and as we collect more information. We may even have backup plans in case the primary plan does not work as expected.

Our research focuses on using UAVs in a swarm to perform SAR missions with the objective of rescuing missing individuals in distress. The main contribution of this paper is the development of a new approach that enables the utilization of drone swarms in search and rescue operations when a probability map of the region is

provided in advance. To achieve this, we have designed a set of heuristics inspired by the work of Giacomossi et al. [6] and Sundelius et al. [4]. Additionally, we have created simulations that can emulate search missions with drones in both 2D and 3D virtual environments. Through these simulations, we evaluate the effectiveness of our approach.

This paper is organized as follows. Section 2 presents the theoretical background. The scenario of interest is presented in Sect. 3. Later, in Sect. 4 we describe our solution. Then we proceed to the experiment's results and discussion in Sect. 5. Lastly, Sect. 6 concludes and shares ideas for future work.

2 Theoretical Background

In this section, we present the main techniques found in the literature for our context.

2.1 Potential Fields

Potential fields have been widely used in the field of aerial robotics for navigation and obstacle avoidance. They provide a way for autonomous aerial robots to generate smooth and safe trajectories by assigning attractive and repulsive forces to the environment. The attractive force pulls the robot towards the goal while the repulsive force repels it from obstacles. The resulting path is a compromise between the two forces that satisfies the objective while avoiding collisions with obstacles. One popular variant of the artificial potential field (APF) method is using the bivariate normal function as a potential field [11]. However, potential fields are known to suffer from local minima and can get stuck in trap situations. To overcome this problem, various enhancements to the basic potential field approach have been proposed, e.g., combining potential fields with other techniques such as machine learning [6]. The potential field that governs the movement of drones within the formation can be defined by.

$$f(x, y) = e^{-\alpha(x-x_c)^2 - \gamma(y-y_c)^2}, \quad (1)$$

where α and γ are constants, $[x_c, y_c]^T$ is the center of the field, and $[x, y]^T$ is the position that is influenced by the potential field. In order to achieve a symmetrical field, both α and γ need to be equal. This ensures the influence to be uniform in all directions, which is important for maintaining coordinated motion within the formation. Specifically, the partial derivatives of (1) create a velocity field that dictates the direction and speed of movement for each drone, based on its position relative to the center of the field. The equation is given by

$$\begin{bmatrix} \frac{\partial f(x,y)}{\partial x} \\ \frac{\partial f(x,y)}{\partial y} \end{bmatrix} = \begin{bmatrix} -2\alpha f(x,y) & 0 \\ 0 & -2\gamma f(x,y) \end{bmatrix} \begin{bmatrix} (x - x_c) \\ (y - y_c) \end{bmatrix}, \quad (2)$$

By modifying α and γ , the potential field can be shaped and scaled to achieve different patterns of drone behavior, allowing for flexibility in the formation's movement.

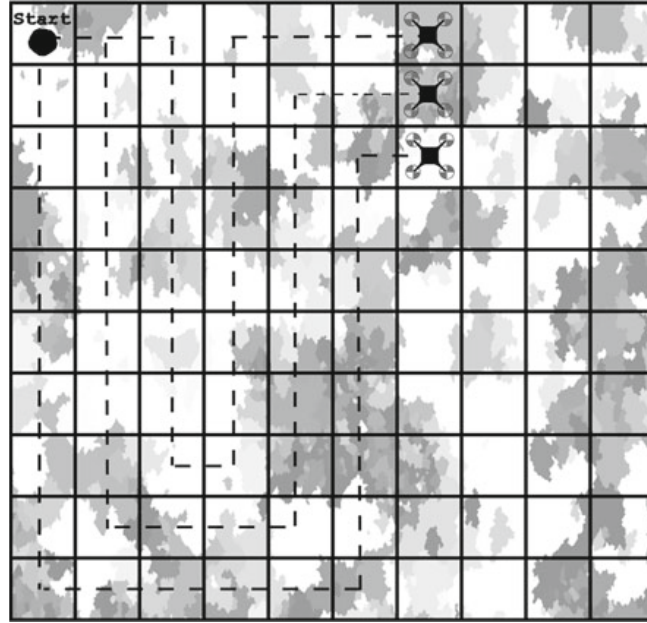
2.2 Search Strategies

There are many approaches to search a specific area, one is providing a probability grid map of where a target might be located, this grid map is divided into cells of equal size. One approach is to use the Bayes Search Algorithm [8] to update the map after each visited cell to a lower probability where nothing is found and update every other cell with a higher probability with the help of an equation. To traverse such a discrete map three techniques are investigated. A baseline lawn mower-like search, that does not take the probabilities into account, a hill climbing approach and an A-star hill climbing mix. Once the search begins, each UAV will select one high probability cell of the map to ensure the distribution of the swarm members in the map, and as soon as the UAV reaches the destination, it will begin the search using one of the search algorithms implemented.

The Lawn Mower Search (LMS) algorithm, also known as row, line or column search, is a search technique that has been used in optimization problems inspired by the motion of a lawn mower. According to the research by Ousingsawat and Earl [12], the LMS divides the search space into subregions and searches in a zigzag pattern, similar to the motion of a lawn mower, exemplified in Fig. 1. This approach is particularly useful for problems with very little information. The LMS has been shown to find optimal solutions, making it a popular choice for many optimization problems [12]. However, LMS has limitations as it can be inefficient and time-consuming. It may require extensive search time in some areas and overlook small, but critical areas. Hence, for more complex scenarios, we need more sophisticated search algorithms that can adapt to different situations and utilize the available information in the most effective way.

The Hill Climbing (HC) algorithm [13] is a powerful optimization technique widely used in computer science and AI. It is a local search algorithm that iteratively moves towards a higher-value solution in the search space, by evaluating neighbouring solutions and selecting the one with the highest value. The HC continues this process until it reaches a peak where no higher-value solution can be found. For convex problems it will always find the optimum solution. However, if the problem is not convex, the algorithm can get stuck in local maxima and minima, resulting in suboptimal solutions. To address this limitation, variants of the HC algorithm have been developed, which introduce randomness and memory to the search process to enable the algorithm to escape local optima and improve the solution. The HC algorithm is useful when there is no information outside the closest surroundings, as

Fig. 1 Example of a swarm formation executing the lawn mower search pattern. The dashed lines show the paths



it follows the most promising path. However, the problem of getting stuck in local maxima remains a challenge. In our previous work [6], we tackled this by proposing a strategy using finite state machines (FSM) to overcome the problem and to enhance the performance of the search.

Lastly, the Bayes Search (BS) algorithm was developed during World War II when the allies were searching for German submarines. It has its origins in the Bayes theorem [8]. The probabilities from the beginning of the search are based on assumptions before any data is collected and are based on experts and sensor data. The first step is to divide the search area into a grid map and assign each cell with a priori-probabilities. The second step is to search the cell with the highest probability, if the target is there, the search is finished. If the target is not there, then the third step is to reduce the probability value of that cell and increase the probability of the other cells correspondingly. The fourth step is to repeat until the target is found. The Bayes formula to update the cells for a search mission can be set up as,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B \cap A)}{P(B)}, \quad (3)$$

where the variables A and B are defined as

$$P(Is\ in\ the\ search\ area|Not\ found) = \frac{P(Is\ in\ the\ search\ area\ AND\ Not\ found)}{P(Not\ found)}, \quad (4)$$

Now let p denote the probability of the target being in a cell and f the probability of that target being found if it is there. Then $P(B)$ and $P(B \cap A)$ can be defined by

$$P(B) = p * (1 - f) + (1 - p), \quad (5)$$

$$P(B \cap A) = p * (1 - f), \quad (6)$$

Now the posterior probability p' of a searched cell can be updated with

$$p' = p \frac{1 - f}{1 - pf}, \quad (7)$$

The posterior probability of other cells can be updated with

$$r' = \frac{r}{1 - pf}, \quad (8)$$

where r is the individual probability of all other cells.

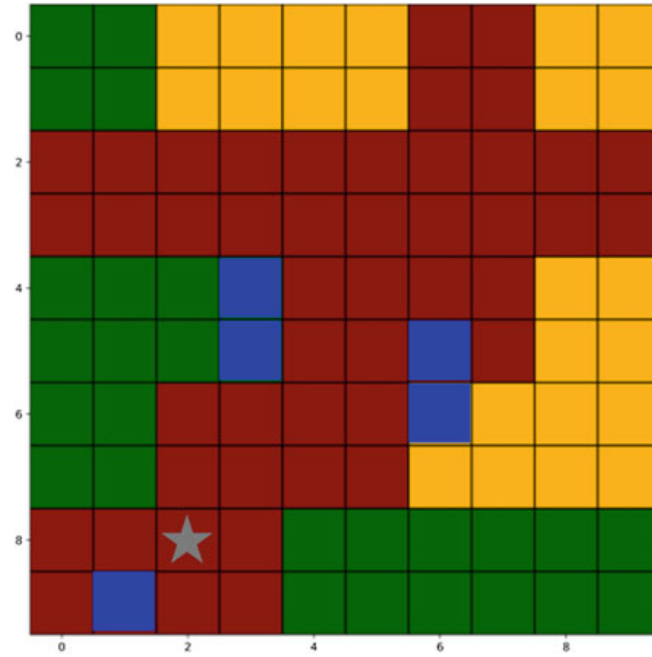
3 Search and Rescue Scenario

In order to optimize patrolling in autonomous scenarios, the task can be broken down into two sub-tasks: low-level and high-level patrolling. Low-level patrolling involves using sensors to detect objects of interest within the area. Meanwhile, high-level patrolling focuses on directing the search towards specific subregions of the area that are more likely to contain the object of interest. This can be accomplished by utilizing additional information, such as previous search patterns and knowledge about the types of objects that are typically found in certain areas. By employing a combination of these tactics, an autonomous system can effectively and efficiently search the patrolling area while minimizing the risk of overlooking crucial objects or areas. Therefore, the task can be broken down into:

- Low-Level Strategy and Tactics for anomaly detection and classification. An efficient patrolling drone is able to identify anomalies and classify them including a number of properties, *e.g.* speed and orientation.
- High-Level Strategy and Tactics for searching the area of interest. An essential part of a patrolling task is to perform the task as efficiently as possible with respect to identifying the target and minimise the risk of missing an object of interest.

Anomaly detection and image recognition are rapidly advancing fields, with potential applications in patrolling. In our previous work [5], we demonstrated this potential by adapting these techniques to patrolling tasks. Building on this foundation, our current work aims to address high-level strategies and tactics for patrolling. We assume a previously provided probability grid map of the search region, as shown in Fig. 2. In this map the region is divided into equally sized cells, where each cell represents the likelihood of finding a person in that sub-region. The probabilities are estimated by a human expert and are updated simultaneously by the drones using

Fig. 2 Example of a probability map. Green, Yellow and Red cells represent low, medium and high probability regions, respectively. No flight zones are illustrated as Blue cells. The star represents the object of interest



search algorithms. To ensure seamless coordination in the swarm, we assume the drones share the same map and are capable of updating the probabilities on the map simultaneously. Furthermore, we assume that the drones are equipped with advanced sensors that can effectively identify the object of interest.

4 Method and Material

Our simulations aim to evaluate collaborative online mission planning algorithms. To compare the efficiency of different algorithms, we limit the scenario to 5 autonomous drones, which is sufficient as proof of concept. We adapted a 2D simulator, as shown in Fig. 3, based on the simulator developed by Giacomossi et al. [6]. The simulation focuses on search missions, where various search algorithms are tested to identify the location of the missing person in the patrolling area and complete the mission.

The simulation environment can be summarized with the diagram in Fig. 4. We implemented a communication model for the swarm to ensure that all UAVs have access to information on which cells have been searched and a basic strategy for searching in a specific direction once all surrounding cells have been visited. The UAVs communicate through a shared map, using the whiteboard model of communication [7] and a centralized ground control system, which enables each drone to read and update map probabilities simultaneously. Additionally, the simulation is connected to the 2D visualization engine that displays the current state of the simulation. By connecting the simulation to a visualization engine, it is possible to monitor and adjust the simulation in real-time. This is useful for testing different scenarios

Fig. 3 2D Simulation view of the search mission in using Bayesian probabilities and HC search

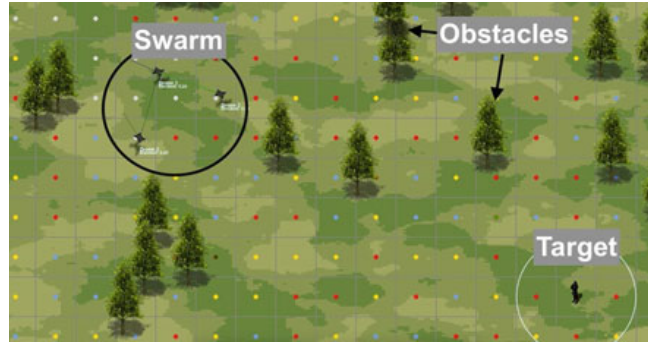
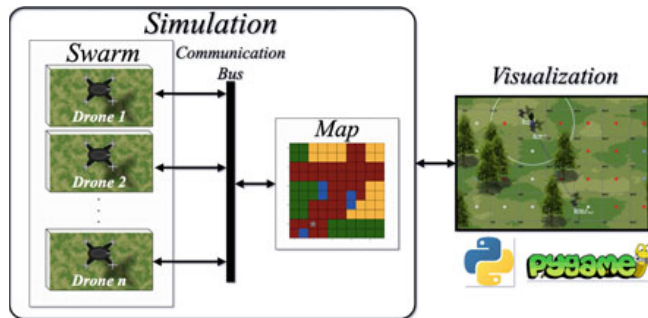


Fig. 4 High-level overview of the components and interactions within the simulation environment



or optimizing the behavior of the drones within the simulation. Additionally, the potential field technique is implemented for collision avoidance and path planning [6].

At the start of the search process, the drones communicate with one another to select different cells on the map with high probabilities and begin the search process, as shown in Fig. 5. Once the drone reaches the intended location, it actively initiates the search process using HC and BS algorithms. In the following Fig. 6, we can observe the search process being performed by the drone. The drone follows a strategy where it visits a cell and, in case the object is not identified, it examines the probabilities of the 8 cells. Then, using the HC algorithm, the drone moves to the cell with the highest probability to continue the search.

In Fig. 7 we illustrate the Bayes algorithm updating the cell probabilities in a simulation. The probabilities of all map cells are updated at each step using the formulas (7) and (8). As seen in frames (a) and (b), the probabilities increase as the drone visits new cells without finding the object.

During each iteration of the simulation, a new probability map is created randomly. The probability for each cell in the map is also randomly assigned. The location of the individual is also updated to a new high probability cell in the newly generated map. Also, new obstacles are randomly positioned in the map, which helps to evaluate the algorithm's performance under different conditions. The random assignment of probabilities to each cell ensures that the simulation results are unbiased towards any particular scenario. For the experiments, we will evaluate the average time taken to

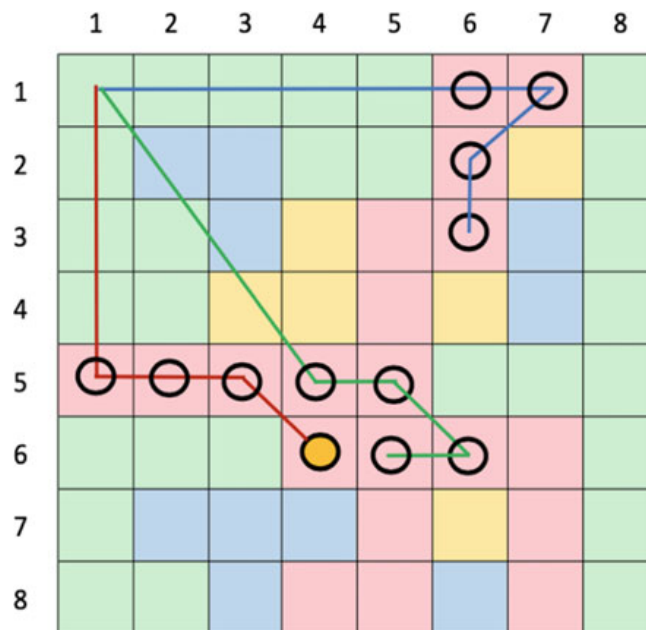


Fig. 5 Example of search process using three drones. Green, yellow, and red cells represent low, medium, and high probability regions, respectively. No flight zones are illustrated as blue cells. The yellow circle represents the object of interest. Each line with a different colour represents the path taken by a different drone

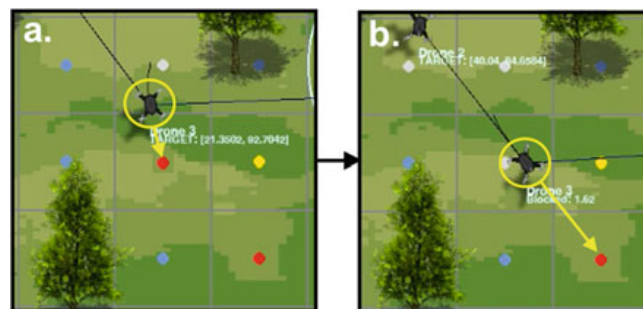


Fig. 6 Drone performing the search in the 8-connected cells. Red, Yellow and Blue cells represent the high, medium and low probabilities respectively. In Gray cells that were already visited

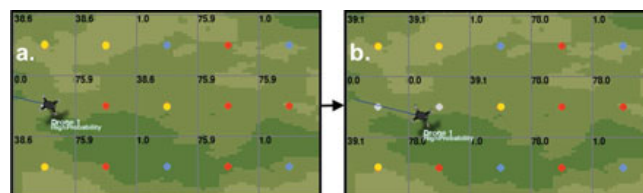


Fig. 7 Drone performs the search using the Bayes algorithm and updates the probabilities associated with each cell as it visits them. These probabilities correspond to the numbers assigned to each cell and are updated at every new cell the drone explores

complete the mission in 270 iterations and compare it to the results obtained in our previous work [6] which employed a strategy based on the lawn mower search.

5 Results and Discussion

In Table 1 we present the results obtained from the experiments. Notably, Experiment 2 demonstrated a shorter mean time of 24.2 s compared to Experiment 1. This indicates that the new strategy utilizing the HC and BS algorithms resulted in an approximately 72% improvement in time efficiency for identifying the missing person. These results suggest that the HC and BS algorithms significantly reduced the time required to complete the task, in comparison to the approach adopted in Experiment 1.

It is important to consider that the drones are being simulated in a 2D environment with reduced degrees of freedom, simplified dynamics, and with idealized identification and communication capabilities. So notice that these results can deviate from real world performance. Furthermore, it demonstrates the importance of improving the search method when the goal is to optimize the time efficiency. Note that the Experiment 2 may be more useful as a model for achieving that goal, but the Experiment 1 is still a safe approach when performing a search when there is no information about the region, *i.e.*, no previous map of probabilities is provided.

The results of the comparison between the mission with and without the aid of a probability map provide evidence that using such a map can significantly reduce search time. An interesting observation from the experiments is that when the target is located in a high probability area with most of its cells in the direct path of the drones, as seen in Fig. 8, the time taken to locate the target is significantly reduced. This is because the drones can fly almost directly to the target, bypassing areas with low probabilities of detection. However, if the target is not in a high probability area, the algorithm may lead the drones to wrong areas, resulting in decreased time efficiency. Therefore, while the use of a probability map can improve search time in optimal scenarios, its effectiveness may vary in situations where the map has incorrect assumptions. Thus, the accuracy of the probability map is a critical factor that has a significant impact on the search.

By leveraging the power of AI and probability mapping, search and rescue operations can become faster, more efficient, and ultimately more successful. Time can be translated into reduction of fuel and energy costs, as well as reduced number of UAVs involved to meet the success criterias of the mission. These results can yield large performance increases by choosing the right mission planning algorithms. When

Table 1 Results of the experiments

Exp	Strategy	Mean Time [s]
1	Lawn Mower - Giacomossi et al. [6]	57.6
2	HC + Bayes	33.4

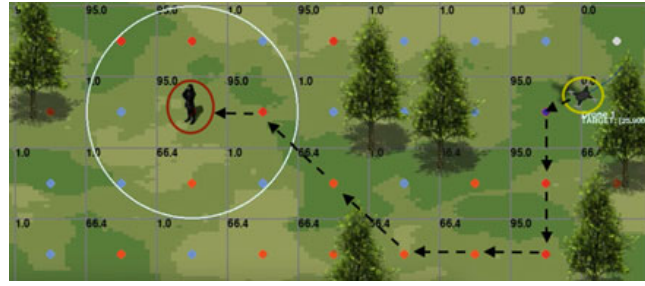


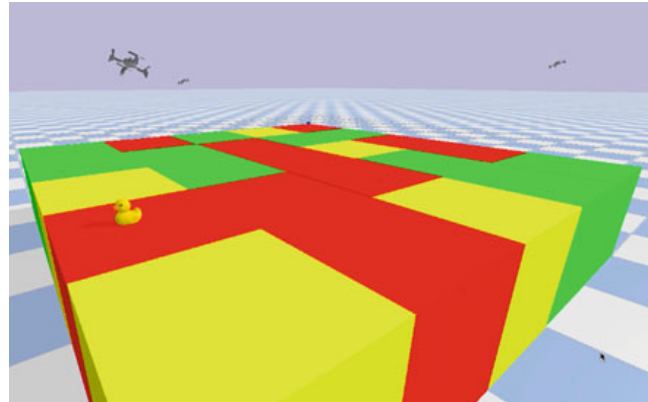
Fig. 8 Drone performing the search in the 8-connected cells. Red, Yellow and Blue cells represent the high, medium and low probabilities respectively. In Gray cells that were already visited. The numbers represent the probability of each cell

performing searches with a map of probabilities and Bayesian updates it is important to make precise assumptions about the probabilities when preparing the map.

6 Conclusion

This work combines AI techniques with a swarm of UAVs to coordinate efficient search and rescue missions in a cooperative and autonomous manner. Moreover, the study demonstrates the potential of AI algorithms for patrolling, target identification, and patrol mission planning. We apply these concepts in a search and rescue scenario, modelled as a testbed for this kind of application where we study strategies and tactics for locating an object of interest *i.e.*, a missing person. The methodologies using search patterns, hill climbing algorithm, bayesian search, and potential fields were merged with the development of simulations to provide an application and evaluation of the concept. In this work we reduced by approximately 72% the time taken in searching for the missing person using the strategies developed in comparison with our previous work, which employed a strategy with no previous information of where the missing person might be located. The findings highlight the importance of continually exploring new approaches and technologies in order to optimize search and rescue missions and increase the chances of saving lives. Also, it is important to note the importance of making precise assumptions when preparing the map, otherwise it may lead to incorrect and not effective searches. For future work, we intend to study the performance of the algorithms in a 3D environment with robust physics and realistic drone dynamics, seen in Fig. 9. Also, the intention in the future is to apply this technique in real drones. Further experimentations with different kinds of maps to evaluate the algorithms and tweaking of the probability update function can be also studied.

Fig. 9 Search and rescue scenario adapted and performed in the py-bullet 3D environment. The cells' colors represent the probability of a subregion of the map. Green refers to low probability cells, Yellow to medium probability and Red cells to high probability cells. The duck represents the missing individual



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